Christian Fauci

Electronic Engineer

(+39) 327-797-1962 christian.fauci2001@gmail.com linkedin.com/christian-fauci github.com/C13v2

About Me

I'm an **Embedded-oriented Electronic Engineer** with a strong focus on **firmware development, real-time systems, and hardware–software integration**. Experienced in **low-level programming and digital signal processing** through hands-on projects involving **control systems and sensor data**. Interested in developing reliable and efficient embedded solutions in industrial, automotive, and semiconductor environments.

Education

University of Palermo

Bachelor of Electronic Engineering — Telecommunications *Expected Graduation: September 2026*

- Courses: Embedded System, Internet of Things, Computer Architecture, Automatic Control, Applied Telecommunications, Digital Signal Processing, Electronic Calculators, Modern Electronics, AutoCAD
- ERASMUS+ — **Univerzita Pardubice**, Czech Republic — Focus on Digital Signal Processing and Signals, Automatic Control, Matlab and Simulink

Technical Skills

Languages: C++, Python, HTML, CSS, JavaScript, Typescript

Libraries/Frameworks: ReactJS, PostgreSQL, NextJS, NodeJS

Projects

Automated Door Control System | Source Code **C++ | Embedded | FSS**

- Designed and implemented a **real-time embedded control** application using a finite state machine **(FSM)** architecture to manage door motion phases with deterministic timing constraints.
- Developed event-driven input handling with sequence validation, timeout monitoring, and safety logic, including **asynchronous STOP** interrupt handling and forced transition to safe system state.
- Implemented software **PWM** generation and multi-timer scheduling to simulate motor control behavior, demonstrating concepts of real-time control, concurrency management, and embedded system timing design.

AVR-based Embedded Password & Alarm System | Source Code **C++ | Embedded | FSM**

- Designed and implemented an **embedded access control system** on AVR microcontroller using a **finite state machine (FSM)** architecture to manage password input, system activation, reset, and alarm states.
- Developed **interrupt-driven input handling** using pin change interrupts (PCINT) for responsive button detection, combined with **timer-based scheduling** (CTC mode) to manage system timing and state transitions.
- Implemented a **security and alarm mechanism** with password validation, error detection, and fail-safe behavior, including buzzer activation, LED signaling, and controlled reset logic for safe system recovery.

Fitness Tracker | Source Code **Python | Pandas | Scikit-learn**

- Designed and implemented a Human Activity Recognition (**HAR**) pipeline from raw IMU accelerometer and gyroscope **signals**, enabling classification of complex motion patterns in real-world conditions.
- Applied **digital signal processing** techniques (**filtering, FFT, windowing**) and feature engineering to improve signal quality and classification performance.
- Developed and evaluated **machine-learning models** (**Random Forest, Neural Networks, KNN, Naive Bayes**), demonstrating integration of **embedded sensing, signal processing, and data-driven modelling** for wearable systems.

Spoken Languages

Italian (Native) English (C1) French (B1) German (A2)